Day 4 — Hydraulics

Open-Channel Flow, Critical Flow, Weirs, Orifices & Blowers

Comprehensive Problem Set

June 20, 2026 • Concept Review + 24 Questions

Today's focus. Uniform flow by Manning's equation; hydraulic radius/depth; specific energy and critical depth; the Froude number and sub/supercritical regimes; hydraulic jumps; flow measurement with weirs and orifices; tank draining; and blower power. **Instructions:** Work each problem on paper, write units at every step, then check the answer key and worked solutions. Use $g = 9.81$ m/s^2 (32.2 ft/s^2) and the NCEES FE Reference Handbook (v10.6).

Concept Review

1. Open-Channel Geometry & Hydraulic Radius

- **Hydraulic radius:** $R_H = \dfrac{A}{P} = \dfrac{\text{cross-sectional area of flow}}{\text{wetted perimeter}}$ — the length scale for non-circular and open conduits.
- **Hydraulic diameter:** $D_H = \dfrac{4A}{P} = 4R_H$. For a circular pipe flowing *full*, $D_H = D$, so $R_H = \dfrac{D}{4}$ (note the factor-of-4 difference from D_H).
- **Top width and hydraulic depth:** T = width of the free surface; $y_h = \dfrac{A}{T}$. The hydraulic depth y_h (not the maximum depth) is used in the Froude number.
- **Rectangular channel** of width b and flow depth y: $A = by$, $P = b + 2y$, $T = b$, $R_H = \dfrac{by}{b+2y}$.

Handbook: Fluid Mechanics — Flow in Noncircular Conduits; Civil Engineering — Open-Channel Flow

2. Manning's Equation (Uniform Flow)

- **Discharge / velocity:** $Q = \dfrac{K}{n} A\, R_H^{2/3} S^{1/2}, \qquad v = \dfrac{K}{n} R_H^{2/3} S^{1/2}$.
- **Unit constant:** $K = 1.486$ (USCS), $K = 1.0$ (SI). n = roughness coefficient; S = channel (energy-grade-line) slope, $S = h_f/L$.
- **Uniform flow:** depth and velocity are constant along the reach, so the bed slope, water-surface slope, and energy slope are equal. This is the condition Manning's equation describes.
- **Sensitivities:** $Q \propto S^{1/2}$ and $Q \propto R_H^{2/3}$; a larger roughness n *lowers* v and Q.

Handbook: Civil Engineering — Manning's Equation

3. Specific Energy & Critical Depth

- **Specific energy** (relative to the channel bottom): $E = \alpha \dfrac{v^2}{2g} + y = \alpha \dfrac{Q^2}{2gA^2} + y$, with kinetic-energy correction $\alpha \approx 1.0$.

- For a given Q, E is *minimum* at the **critical depth** y_c. Two depths with the same E are **alternate depths** (one sub-, one supercritical).
- **General critical condition:** $\dfrac{Q^2}{g} = \dfrac{A^3}{T}$.
- **Rectangular channel:** $y_c = \left(\dfrac{q^2}{g}\right)^{1/3}$, where unit discharge $q = \dfrac{Q}{B}$ and B = channel width.
- Background (rectangular only, not stated explicitly in the Handbook): $E_{\min} = \frac{3}{2} y_c$.

Handbook: Civil Engineering — Specific Energy; Critical Depth

4. Froude Number & Flow Regimes

- **Froude number** (inertia/gravity): $Fr = \dfrac{v}{\sqrt{g\, y_h}} = \sqrt{\dfrac{Q^2 T}{g A^3}}$, with $y_h = A/T$.
- **Subcritical** $Fr < 1$ (deep, slow; disturbances travel upstream, control is downstream). **Critical** $Fr = 1$. **Supercritical** $Fr > 1$ (shallow, fast; control is upstream).

Handbook: Civil Engineering — Froude number

5. Hydraulic Jump

- An abrupt **supercritical → subcritical** transition that *dissipates* energy (e.g., below a spillway or sluice gate).
- **Sequent (conjugate) depths:** $y_2 = \dfrac{y_1}{2}\left(-1 + \sqrt{1 + 8\, Fr_1^2}\right)$, where y_1 is the upstream supercritical depth.
- Momentum function $M = \dfrac{Q^2}{gA} + A\, h_c$ (h_c = depth to the centroid) is conserved across the jump; sequent depths share the same M.

Handbook: Civil Engineering — Hydraulic Jump; Momentum Depth Diagram

6. Weirs (Flow Measurement)

- **Rectangular, suppressed** (no end contractions): $Q = CLH^{3/2}$.
- **Rectangular, contracted**: $Q = C(L - 0.2H)H^{3/2}$ — end contractions reduce the effective crest length.
- **90° V-notch**: $Q = CH^{5/2}$.
- **Coefficients:** rectangular $C = 3.33$ (USCS) / 1.84 (SI); 90° V-notch $C = 2.54$ (USCS) / 1.40 (SI). L = crest length, H = head over the crest.

Handbook: Civil Engineering — Weir Formulas

7. Orifices (Flow Measurement & Tank Draining)

- **Free discharge to atmosphere:** $Q = CA_0\sqrt{2gh}$ (h measured to the centroid of the opening).
- **Submerged orifice:** $Q = CA\sqrt{2g(h_1 - h_2)}$ ($h_1 - h_2$ = head difference between the two surfaces).
- **Discharge coefficient:** $C = C_cC_v$ (contraction × velocity). Sharp-edged $C \approx 0.61$ ($C_c \approx 0.62$, $C_v \approx 0.98$).
- **Orifice meter (pipe):** $Q = CA_0\sqrt{2g\left(\frac{P_1}{\gamma} + z_1 - \frac{P_2}{\gamma} - z_2\right)}$, $C = \dfrac{C_vC_c}{\sqrt{1 - C_c^2(A_0/A_1)^2}}$.
- **Time to drain a tank:** $\Delta t = \dfrac{2(A_t/A_0)}{\sqrt{2g}}\left(h_1^{1/2} - h_2^{1/2}\right)$, with $A_t = \dfrac{\pi D_t^2}{4}$.

Handbook: Fluid Mechanics — Orifice Discharging Freely into Atmosphere; Submerged Orifice; Orifices

8. Blowers

- **Power requirement:** $P_w = \dfrac{WRT_1}{C\,n\,e}\left[\left(\dfrac{P_2}{P_1}\right)^{0.283} - 1\right]$.
- C = 550 ft-lbf/(sec-hp) (USCS) or 29.7 (SI); W = weight flow of air (lb/sec); R = 53.3 ft-lbf/(lb air-°R); T_1 = absolute inlet temperature (°R); P_1, P_2 = absolute inlet/outlet pressures; $n = (k-1)/k = 0.283$ for air; e = efficiency (typically 0.70–0.90).

Handbook: Fluid Mechanics — Blowers

9. Hazen–Williams & Flumes (notes)

- **Hazen–Williams** (water, turbulent) also appears for open channels: $Q = k_1CAR_H^{0.63}S^{0.54}$, k_1 = 1.318 (USCS) / 0.849 (SI); higher C = smoother.
- **Flumes** (e.g., Parshall) measure flow by forcing critical flow at a throat. The FE Handbook gives *no* flume rating equation, so any FE flume item is conceptual — never compute a flume rating from a memorized formula. *(not in Handbook)*

Handbook: Civil Engineering — Hazen-Williams Equation

10. Partial-Flow Circular Sewers, Culverts & Channel Sections

- **Hydraulic-element (partial-flow) graph for circular sewers** (Handbook): plots the ratios $\frac{v}{v_f}$, $\frac{Q}{Q_f}$, $\frac{A}{A_f}$, $\frac{R}{R_f}$ against relative depth $\frac{d}{D}$ ("f" = full-pipe value). Use it for a partly full circular pipe instead of recomputing R_H by hand.
- Key readings from that curve: maximum *velocity* occurs near $d/D \approx 0.81$ and maximum *discharge* near $d/D \approx 0.93$ — a pipe carries slightly *more* than its full-flow Q just before flowing full.
- **Trapezoidal channel** (bottom width b, side slope z:1 = horizontal:vertical, depth y): $A = (b + zy)y$, $P = b + 2y\sqrt{1+z^2}$, $T = b + 2zy$. Combine with Manning and $R_H = A/P$.
- **Culverts.** *(Not in the FE Handbook — background only.)* Analyzed with energy and continuity: inlet control behaves like an orifice or weir, outlet control like full-pipe flow with friction and minor losses. Solve any FE culvert item with Manning, the energy equation, or the orifice relation — never a memorized culvert nomograph.

Handbook: Civil Engineering — Hydraulic-Element (Partial Flow) Graph for Circular Sewers

Questions

1. A rectangular open channel is 3 m wide and carries water at a flow depth of 1 m. The hydraulic radius R_H is most nearly:

   (A) 0.50 m
   (B) 0.60 m
   (C) 1.00 m
   (D) 3.00 m

2. A circular storm sewer of diameter D flows completely full. Its hydraulic radius equals:

   (A) $D/2$
   (B) D
   (C) $D/4$
   (D) $D/8$

3. A rectangular channel ($b = 3$ m, $y = 1$ m) has $n = 0.013$ and bed slope $S = 0.001$. The uniform-flow velocity (SI Manning) is most nearly:

   (A) 0.86 m/s
   (B) 1.20 m/s
   (C) 1.73 m/s
   (D) 2.05 m/s

4. A rectangular channel 4 m wide flows 1.5 m deep with $n = 0.015$ and $S = 0.0009$. The discharge (SI Manning) is most nearly:

   (A) 6.5 m^3/s
   (B) 8.9 m^3/s
   (C) 13.4 m^3/s
   (D) 10.8 m^3/s

5. In Manning's equation, the unit constant K for USCS units is:

   (A) 1.0
   (B) 1.318
   (C) 1.486
   (D) 0.849

6. Holding discharge, geometry, and slope constant, increasing the roughness coefficient n causes the uniform-flow velocity to:

   (A) increase
   (B) decrease
   (C) stay the same
   (D) double

7. A rectangular channel carries a unit discharge $q = 2.5$ m^2/s at a depth of 1.5 m. The specific energy ($\alpha = 1.0$) is most nearly:

   (A) 1.64 m
   (B) 1.50 m
   (C) 1.78 m
   (D) 2.00 m

8. A rectangular channel 4 m wide carries $Q = 10$ m^3/s. The critical depth is most nearly:

   (A) 0.61 m
   (B) 1.10 m
   (C) 0.86 m
   (D) 1.25 m

9. At critical flow in an open channel:

   (A) specific energy is maximum and $Fr = 0$
   (B) specific energy is minimum and $Fr = 1$
   (C) the velocity is zero
   (D) the depth is maximum

10. Water flows 0.5 m deep at 3 m/s in a wide rectangular channel. The Froude number and regime are most nearly:

(A) 0.74, subcritical
(B) 1.35, supercritical
(C) 1.35, subcritical
(D) 0.74, supercritical

11. A flow with a Froude number of 0.6 is classified as:

(A) supercritical
(B) critical
(C) subcritical
(D) turbulent

12. "Alternate depths" are two different flow depths that share the same:

(A) discharge only
(B) specific energy
(C) momentum
(D) velocity

13. A hydraulic jump forms where the upstream depth is 0.4 m and $Fr_1 = 2.5$. The downstream (sequent) depth is most nearly:

(A) 0.80 m
(B) 1.00 m
(C) 1.23 m
(D) 1.60 m

14. A hydraulic jump occurs when flow transitions from:

(A) subcritical to supercritical, gaining energy
(B) supercritical to subcritical, dissipating energy
(C) laminar to turbulent flow
(D) uniform to nonuniform with no energy loss

15. A suppressed rectangular weir 5 ft long discharges under a head of 1 ft ($C = 3.33$, USCS). The discharge is most nearly:

(A) 11.1 cfs
(B) 22.0 cfs
(C) 33.3 cfs
(D) 16.7 cfs

16. A 90° V-notch weir operates under a head of 0.8 ft ($C = 2.54$, USCS). The discharge is most nearly:

(A) 0.91 cfs
(B) 2.03 cfs
(C) 1.45 cfs
(D) 2.54 cfs

17. **(Numeric entry)** A suppressed rectangular weir 2 m long operates under a head of 0.3 m. Using the SI weir coefficient $C = 1.84$, determine the discharge Q (in m^3/s).

18. The discharge over a *contracted* rectangular weir is given by:

(A) $Q = CLH^{3/2}$
(B) $Q = C(L - 0.2H)H^{3/2}$
(C) $Q = CH^{5/2}$
(D) $Q = CA\sqrt{2gh}$

19. A sharp-edged orifice ($d = 0.10$ m, $C = 0.61$) discharges freely with 2 m of head above its center. The discharge is most nearly:

(A) 0.015 m^3/s
(B) 0.049 m^3/s
(C) 0.060 m^3/s
(D) 0.030 m^3/s

20. A submerged orifice ($A = 0.05$ m^2, $C = 0.61$) operates under a 1.5 m difference between the upstream and downstream water surfaces. The discharge is most nearly:

(A) 0.165 m^3/s
(B) 0.083 m^3/s
(C) 0.117 m^3/s
(D) 0.235 m^3/s

21. **(Numeric entry)** A cylindrical tank 2 m in diameter drains through an orifice of area 0.01 m^2 in its base. Using the Handbook tank-draining relation (no discharge coefficient), find the time (in seconds) for the water level to fall from 4 m to 1 m above the orifice.

22. A blower moves air ($W = 2$ lb/sec, $R = 53.3$ ft-lbf/lb-°R) at an inlet temperature of 530 °R from $P_1 = 14.7$ psia to $P_2 = 18.7$ psia at efficiency $e = 0.80$. Using the Handbook blower equation ($C = 550$, $n = 0.283$), the power requirement is most nearly:

(A) 18 hp
(B) 32 hp
(C) 25 hp
(D) 45 hp

23. According to the Handbook's hydraulic-element (partial-flow) graph for circular sewers, the *maximum discharge* in a circular pipe occurs at a relative depth d/D of approximately:

(A) 0.50
(B) 0.71
(C) 0.93
(D) 1.00

24. A trapezoidal channel has bottom width 3 m, side slopes 2H:1V ($z = 2$), and flows 1 m deep with $n = 0.025$ and $S = 0.0016$. The discharge (SI Manning) is most nearly:

(A) 4.3 m^3/s
(B) 6.1 m^3/s
(C) 7.8 m^3/s
(D) 9.5 m^3/s

Answer Key

1. B	**2.** C	**3.** C	**4.** D	**5.** C	**6.** B
7. A	**8.** C	**9.** B	**10.** B	**11.** C	**12.** B
13. C	**14.** B	**15.** D	**16.** C	**17.** 0.60 m^3/s	**18.** B
19. D	**20.** A	**21.** ≈142 s	**22.** B	**23.** C	**24.** B

Worked Solutions

1. $A = by = 3(1) = 3\ \text{m}^2$; $P = b + 2y = 3 + 2(1) = 5$ m; $R_H = A/P = 3/5 = \mathbf{0.60\ m}$.

   Handbook: Flow in Noncircular Conduits

2. For any circular conduit flowing full, $A = \frac{\pi}{4}D^2$ and $P = \pi D$, so $R_H = \dfrac{\pi D^2/4}{\pi D} = \dfrac{D}{4}$. **(C)**

   Handbook: Flow in Noncircular Conduits

3. $R_H = \dfrac{3(1)}{3+2} = 0.60$ m. $v = \dfrac{1.0}{0.013}(0.60)^{2/3}(0.001)^{1/2} = 76.9(0.711)(0.0316) = \mathbf{1.73\ m/s}$.

   Handbook: Manning's Equation

4. $A = 4(1.5) = 6\ \text{m}^2$, $P = 4 + 2(1.5) = 7$ m, $R_H = 6/7 = 0.857$ m. $v = \dfrac{1.0}{0.015}(0.857)^{2/3}(0.0009)^{1/2} = 66.7(0.902)(0.03) = 1.80$ m/s. $Q = vA = 1.80(6) = \mathbf{10.8\ m^3/s}$.

   Handbook: Manning's Equation

5. By definition in the Handbook, $K = 1.486$ for USCS units and 1.0 for SI units. **(C)**

   Handbook: Manning's Equation

6. Manning gives $v = \dfrac{K}{n}R_H^{2/3}S^{1/2}$; $v \propto 1/n$, so larger n *decreases* velocity. **(B)**

   Handbook: Manning's Equation

7. $v = q/y = 2.5/1.5 = 1.667$ m/s. $E = y + \dfrac{v^2}{2g} = 1.5 + \dfrac{1.667^2}{2(9.81)} = 1.5 + 0.142 = \mathbf{1.64\ m}$.

   Handbook: Specific Energy

8. $q = Q/B = 10/4 = 2.5\ \text{m}^2/\text{s}$. $y_c = \left(\dfrac{q^2}{g}\right)^{1/3} = \left(\dfrac{2.5^2}{9.81}\right)^{1/3} = (0.637)^{1/3} = \mathbf{0.86\ m}$.

   Handbook: Critical Depth

9. Critical flow is the state of *minimum specific energy* for a given discharge, where $Fr = 1$. **(B)**

   Handbook: Critical Depth; Froude number

10. For a wide rectangular channel $y_h = y$, so $Fr = \dfrac{v}{\sqrt{gy}} = \dfrac{3}{\sqrt{9.81(0.5)}} = \dfrac{3}{2.215} = 1.35 > 1 \Rightarrow$ supercritical. **(B)**

Handbook: Froude number

11. $Fr = 0.6 < 1$, so the flow is subcritical. **(C)**

Handbook: Froude number

12. Alternate depths are the two depths (one sub-, one supercritical) that yield the *same specific energy* for a given Q. **(B)**

Handbook: Specific Energy Diagram

13. $y_2 = \dfrac{y_1}{2}\left(-1 + \sqrt{1 + 8Fr_1^2}\right) = \dfrac{0.4}{2}\left(-1 + \sqrt{1 + 8(2.5)^2}\right) = 0.2(-1 + \sqrt{51}) = 0.2(6.141) = \mathbf{1.23\ m}$.

Handbook: Hydraulic Jump

14. A hydraulic jump is a sudden supercritical-to-subcritical transition that *dissipates* mechanical energy. **(B)**

Handbook: Hydraulic Jump

15. Suppressed rectangular weir: $Q = CLH^{3/2} = 3.33(5)(1)^{3/2} = \mathbf{16.7\ cfs}$. **(D)**

Handbook: Weir Formulas

16. 90° V-notch: $Q = CH^{5/2} = 2.54(0.8)^{5/2} = 2.54(0.572) = \mathbf{1.45\ cfs}$. **(C)**

Handbook: Weir Formulas

17. Suppressed rectangular weir (SI): $Q = CLH^{3/2} = 1.84(2)(0.3)^{3/2} = 1.84(2)(0.164) = \mathbf{0.60\ m^3/s}$.

Handbook: Weir Formulas

18. End contractions shorten the effective crest, giving $Q = C(L - 0.2H)H^{3/2}$. **(B)**

Handbook: Weir Formulas

19. $A_0 = \dfrac{\pi}{4}(0.10)^2 = 7.85 \times 10^{-3}\ \text{m}^2$. $Q = CA_0\sqrt{2gh} = 0.61(7.85 \times 10^{-3})\sqrt{2(9.81)(2)} = 0.61(7.85 \times 10^{-3})(6.26) = \mathbf{0.030\ m^3/s}$. **(D)**

Handbook: Orifice Discharging Freely into Atmosphere

20. Submerged orifice: $Q = CA\sqrt{2g(h_1 - h_2)} = 0.61(0.05)\sqrt{2(9.81)(1.5)} = \mathbf{0.165\ m^3/s}$. **(A)**

Handbook: Submerged Orifice Operating under Steady-Flow Conditions

21. $A_t = \dfrac{\pi}{4}(2)^2 = 3.142\ \text{m}^2$. $\Delta t = \dfrac{2(A_t/A_0)}{\sqrt{2g}}\left(h_1^{1/2} - h_2^{1/2}\right) = \dfrac{2(3.142/0.01)}{\sqrt{2(9.81)}}\left(\sqrt{4} - \sqrt{1}\right) = \dfrac{628.3}{4.429}(1) = \mathbf{142\ s}$.

Handbook: Orifice Discharging Freely into Atmosphere (time to drain a tank)

22. $P_w = \dfrac{WRT_1}{Cne}\left[\left(\dfrac{P_2}{P_1}\right)^{0.283} - 1\right] = \dfrac{2(53.3)(530)}{550(0.283)(0.80)}\left[\left(\dfrac{18.7}{14.7}\right)^{0.283} - 1\right] = 453.7(0.0705) = \mathbf{32\ hp}$. **(B)**

Handbook: Blowers

23. From the hydraulic-element graph, the discharge ratio Q/Q_f peaks at a relative depth of about $d/D \approx 0.93$ (slightly above the full-flow value), not at $d/D = 1.0$. **(C)**

Handbook: Hydraulic-Element (Partial Flow) Graph for Circular Sewers

24. Trapezoid: $A = (b + zy)y = (3 + 2 \cdot 1)(1) = 5$ m^2; $P = b + 2y\sqrt{1 + z^2} = 3 + 2\sqrt{5} = 7.47$ m; $R_H = 5/7.47 = 0.669$ m. $v = \dfrac{1.0}{0.025}(0.669)^{2/3}(0.0016)^{1/2} = 40(0.765)(0.04) = 1.22$ m/s. $Q = vA = 1.22(5) =$ **6.1 m^3/s. (B)**

Handbook: Manning's Equation